

THE BEST BOOK I HAVE EVER READ!

My favorite time is spent in bookstores, where I aimlessly move from book to book in an attempt to make a decision as to whether to invest the time in reading it. My buying is frequently made on impulse, based on superficial but suggestive clues. Frequently I have nothing but a book jacket as appendage to my decision making. Jackets often contain praise by someone, famous or not, or excerpts from a book review. Good praise by a famous and respected person or a well-known magazine would sway me into buying the book.

What is the problem? I tend to confuse a book review, which is supposed to be an assessment of the quality of the book, with the *best* book reviews, marred with the same survivorship biases. I mistake the distribution of the maximum of a variable with that of the variable itself. The publisher will never put on the jacket of the book anything but the best praise. Some authors go even a step beyond, taking a tepid or even unfavorable book review and selecting words in it that appear to praise the book. One such example came from one Paul Wilmott (an English financial mathematician of rare brilliance and irreverence) who managed to announce that I gave him his "first bad review", yet used excerpts from it as praise on the book jacket (we later became friends, which allowed me to extract an endorsement from him).

The first time I was fooled by this bias was upon buying, when I was 16, *Manhattan Transfer*, a book by John Dos Passos, the American writer, based on praise on the jacket by the philosopher Jean-Paul Sartre, who claimed something to the effect that Dos Passos was the greatest writer of our time. This simple remark, possibly blurted out in a state of intoxication or extreme enthusiasm, caused Dos Passos to become required reading in European intellectual circles, as Sartre's remark was mistaken for a consensus estimate of the quality of Dos Passos rather than what it was, the best remark. (In spite of having received the Nobel Prize in literature, Dos Passos has reverted to obscurity.)

THE BACKTESTER

A programmer helped me build a *backtester*. It is a software program connected to a database of historical prices which allows me to check

the hypothetical past performance of any trading rule of average complexity. I can just apply a mechanical trading rule, like buy NASDAQ stocks if they close more than 1.83% above their average of the previous week, and immediately get an idea of its past performance. The screen will flash my hypothetical track record associated with the trading rule. If I do not like the results, I can change the percentage, to say, 1.2%. I can also make the rule more complex. I will keep trying until I find something that works well.

What am I doing? The exact same task of looking for the survivor within the set of rules that can possibly work. I am *fitting* the rule on the data. This activity is called *data snooping*. The more I try, the more I am likely, by mere luck, to find a rule that worked on past data. A random series will always present some detectable pattern. I am convinced that there exists a tradable security in the Western world that would be 100% correlated with the changes in temperature in Oulan Bator, Mongolia.

To get technical, there are even worse extensions. An outstanding recent paper by Sullivan, Timmerman and White⁸ goes further and considers that the rules that may be in use successfully today may be the result of a survivorship bias.

Suppose that, over time, investors have experimented with technical trading rules drawn from a very wide universe – in principle thousands of parameterizations of a variety of types of rules. As time progresses, the rules that happen to perform well historically receive more attention and are considered "serious contenders" by the investment community, while unsuccessful trading rules are more likely to be forgotten. ... If enough trading rules are considered over time, some rules are bound by pure luck, even in a very large sample, to produce superior performance even if they do not genuinely possess predictive power over asset returns. Of course, inference based solely on the subset of surviving trading rules may be misleading in this context since it does not account for the full set of initial trading rules, most of which are unlikely to have underperformed.

I have to decry some excesses in back-testing that I have closely witnessed in my private career. There is an excellent product designed